

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Medium

Question

The sum of a number x and 7 is twice as large as a number y . The number y is 3 less than the number x . Which system of equations describes this situation?

Answer

A. $x + 7 = 2y$

$y = x - 3$

B. $x + 7 = 2y$

$y = 3 - x$

C. $2(x + 7) = y$

$y = x - 3$

D. $2(x + 7) = y$

$y = 3 - x$

Correct Answer: A

Rationale

Choice A is correct. It's given that the sum of a number x and 7 is twice as large as a number y . This can be described by the equation $x + 7 = 2y$. It's also given that the number y is 3 less than the number x . This can be described by the equation $y = x - 3$. Therefore, the system consisting of the equations $x + 7 = 2y$ and $y = x - 3$ describes this situation.

Choice B is incorrect. The equation $y = 3 - x$ describes a situation where the number y is x less than 3 .

Choice C is incorrect. The equation $2(x + 7) = y$ describes a situation where the number y is twice the sum of a number x and 7 .

Choice D is incorrect. The equation $2(x + 7) = y$ describes a situation where the number y is twice the sum of a number x and 7 , and the equation $y = 3 - x$ describes a situation where a number y is x less than 3 .